

HomeLoanAdvisor:

A ConsumerFacing Mortgage Risk Intelligence App

Jayden Jiang, Ana Collado, Ariba Razzaq, Sabina Baraili

Introduction

Mortgage risk assessment is typically designed for lenders rather than borrowers. As a result, many prospective homebuyers make major financial decisions without access to clear, data-driven guidance about their own repayment risk. This project addresses that gap by developing Home Loan Advisor, an interactive application that estimates mortgage delinquency risk from borrower and loan characteristics and presents the result in an interpretable format for the consumer. The project combines historical Freddie Mac loan-level data, machine learning, and a deployed R Shiny interface to create a consumer decision support tool.

Problem and Motivation

Banks use structured underwriting systems to evaluate repayment risk, but consumers typically do not have an equivalent tool for understanding how variables such as credit score, debt-to-income ratio, loan-to-value ratio, and interest rate may affect their own mortgage risk. The purpose of this project was therefore to translate historical mortgage origination and performance data into a more accessible application that allows users to explore risk before signing a loan.

Data and Methods

The project utilizes the Freddie Mac Single Family Loan-Level Datasets (which are released quarterly) and formulates the task as a binary classification problem: whether a loan ever becomes 90+ days delinquent. The baseline workflow used a 150,000 loan sample with a 4.31% positive class rate, which makes the problem strongly imbalanced. Preprocessing included target construction, placeholder handling, and encoding of categorical variables and one-hot. The baseline model was a logistic regression model, implemented in the tidymodels framework, and then deployed through R Shiny.

The **final** workflow utilized both Freddie Mac "Origination" and "Performance" files from 2018 to 2022 (40 datasets). This process produced a training dataset of 8,746,063 loans and a testing dataset of 3,748,322 loans.

Preprocessing for the final workflow included date feature engineering, categorical encoding, numeric imputation, target construction, and output stratified train/test splits by quarter and delinquency status.

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Risk tiers were created to ease communication with the consumer through the app. We compared Logistic Regression, Random Forest and Histogram Gradient Boosting models. The models were evaluated using ROC-AUC, precision, recall, and F1 score. Due to the imbalance of the delinquency class ROC may not be the best evaluation metric for classification performance. Several design decisions were important for the final application. Income is shown for affordability context only and is not treated as a core Freddie Mac model feature. The evaluation threshold was set at 0.10 to improve interoperability under class imbalance.

Results

The logistic-regression baseline achieved a mean cross-validated AUC of 0.753 and a held-out test AUC of 0.755. These results indicate that the model can distinguish riskier loans from safer loans substantially better than random chance, while also remaining stable across validation folds. Because the positive class rate is only 4.31%, the project emphasizes AUC rather than raw accuracy as the main performance metric. The system also includes confusion-matrix output, calibration evidence, and borrower demo cases to support interpretation of model behavior.

Application Design

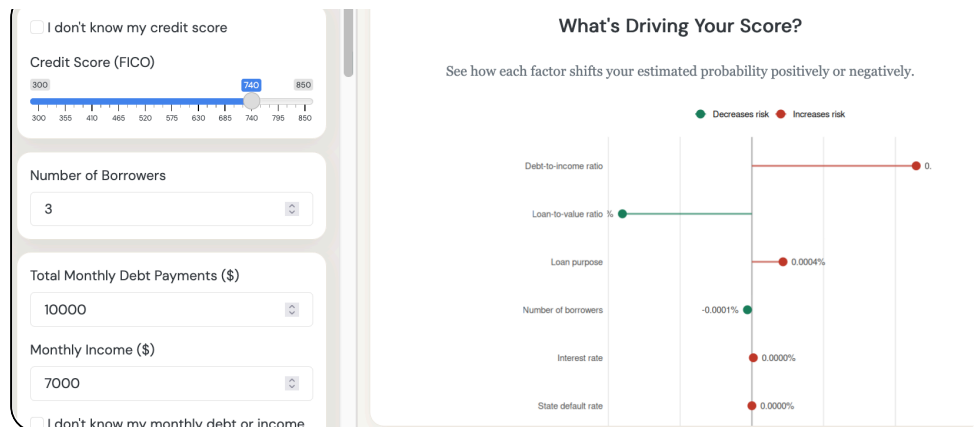
The final project extends beyond a static modeling report. The application is customer facing. A borrower can input their information into fields and receive outputs including scenario simulation, sensitivity analysis, and model-evidence sections. It also supports missing-value handling through “I don’t know” options, which substitute dataset medians and provide a visible warning to the user. In addition to logistic regression, the final system includes Random Forest so that users can compare a linear baseline model with a tree-based model that is better suited to representing non-linear relationships and interaction effects.

The screenshot displays the 'The Home Loan Advisor' app interface. At the top, there's a navigation bar with 'My Profile', 'Scenarios', 'Sensitivity', and 'About the Models'. The main content is divided into two panels. The left panel, titled 'Your Financial Profile', contains a form with a dropdown for 'Property State' (set to 'DC') and a 'Credit Score (FICO)' slider (set to 740). The right panel, titled 'Your Risk Assessment', shows the 'Estimated probability of reaching 90+ days delinquent' as '0.005%' (Fewer than 1 in 10,000) with a 'Lower risk' status bar. Below this, a section titled 'What's Driving Your Score?' is partially visible.

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Significance

The main contribution of this project is that it turns a predictive modeling workflow into a user facing application. Instead of ending with a notebook analysis, the project delivers deployed software that allows users to interact with the model directly. From a data science perspective, this demonstrates the full workflow of problem formulation, data preparation, modeling, evaluation, and deployment. From a practical perspective, it offers a simplified way for borrowers to think about mortgage repayment risk before making a major financial commitment.

Limitations

The models are based on public Freddie Mac data rather than a lender's full internal underwriting process. The data reflects Freddie Mac's own requirements and as such excludes populations (veterans for example). The displayed risk tiers are not calibrated business rules, and income is included for context rather than as a direct training feature. The final demo materials also identify future extensions, including XGBoost, a broader data window, usability testing, and more formal calibration of risk thresholds.

Conclusion

Home Loan Advisor demonstrates that historical mortgage data can support a borrower facing machine learning application. Using Freddie Mac loan data, the project built a serious delinquency prediction workflow and translated it into a deployed Shiny app with scenario analysis, sensitivity analysis, and transparent model evidence. The final result is shows the predictive task is possible for the benefit of a consumer.

